

Curriculum vitae - 27/10/2025

Personal Information and Contact

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PhD Project Description

My PhD project focuses on investigating the functionality of smithRNAs, a class of miRNA-like small RNAs encoded in mitochondria that appear to influence nuclear transcription through RNA interference. Additionally, I am working on quantifying tissue-specific smithRNA expression levels in the Manila clam (*Ruditapes philippinarum*). These objectives are part of the nationally relevant PRIN2020 project, *MitoMicro – Mitochondrial MicroRNAs, a new perspective on mitochondrial function and their role in eukaryotic evolution*, led by Marco Passamonti (PI).

Overall, my research interests encompass the broad theme of mito-nuclear coevolution in metazoans. In particular, I am focusing on the impact of mitonuclear coevolution in shaping the phylogenetic signal.

Furthermore, my collaboration with the University of Modena and Reggio Emilia (PI: Prof. Antonio M. Todaro) on mitochondrial DNA in Gastrotricha has led to several side projects on this small phylum, most notably its first large-scale phylogenomic study.

Technical Skills

Programming: Proficient in Bash and R. Experience with Python.

Strong expertise in conducting phylogenomic analyses and managing large-scale genomic data.

Wet Lab: DNA and RNA extractions, PCR, RT-PCR.

Education

Master's Degree in Biodiversity and Evolution

University of Bologna, Oct 2021 – Oct 2023

Final grade: 110/110 cum laude

Thesis: "*The phylogenetic analysis based on nuclear OXPHOS genes reveals clues of mitonuclear coevolution in Squamata*" - Supervisor: Dr. Federico Plazzi

Bachelor's Degree in Biology

University of Bologna, Oct 2018 – Oct 2021

Final grade: 102/110

Thesis: "*Population density of the solitary azooxanthellate coral Caryophyllia inornata along an environmental gradient generated by CO₂ vent*" - Supervisor: Prof. Stefano Goffredo

Partecipation in Competitive Projects

PRIN 2020: MitoMicro - Mitochondrial MicroRNAs, a new perspective for mitochondrial function and their role in Eukaryotic evolution

Program funded to Marco Passamonti by the Italian Ministry of Education, University and Research (MIUR). Role: Investigation.

Publications

Wallnoefer O, Formaggioni A, Plazzi F, Passamonti M. *Convergent evolution in nuclear and mitochondrial OXPHOS subunits underlies the phylogenetic discordance in deep lineages of Squamata*. Molecular Phylogenetics and Evolution. 2025 Apr 14;208:108358. doi: 10.1016/j.ympev.2025.108358

Conference Oral Presentations

- 3rd Joint Congress on Evolutionary Biology - Virtual Conference; 27-28 June 2024
- 83rd Congress of the Italian Zoological Union - Pisa (IT); 11-14 September 2024
- Evolution 2025 - Virtual Conference; 29-30 May 2025

- Willi Hennig Society 2025 - Hong Kong; 22-25 July
- SMBE 2025 - Beijing (but remote attendance); 20-24 July

Poster Presentations

- Annual Meeting of the Society for Molecular Biology and Evolution (SMBE) - Puerto Vallarta (MX); 07-11 July 2024
- SIBE Congress 2024 - Naples (IT); 08-11 September 2024

Main Courses and Seminars

- *Fundamental and Applied Research* (Annalisa Pelizza - FILCOM, Unibo; Fabrizio Ghiselli - BiGeA; 12/2023)
- *Statistical Course* (Stefania Sarno - BiGeA; Federico Plazzi - BiGeA; 22-26/01/2024)
- *Virtual Workshop on Evolutionary Rate Covariation (ERC)* (Prof. Evan Forsythe, Prof. Dan Sloan; 08/08/2024)
- Scientific Writing and Communication Course (Gabor Lovei - Aarhus University; 04-08/11/2024)
- *Raman Laboratory Course* (Davide Cavalletti - BiGeA; 16/01/2025)
- Inferring phylogenies from metagenomic data (SIBE; Omar Rota-Stabelli; 08/09/2024)

Grants

- *Kurt Milton Pickett Intercontinental Award* - The Willy Hennig Society; awarded for participation in XLII Annual Meeting of the Willi Hennig Society, Hong Kong (scheduled for 06/2025)

Academic activities

- Organizer of the seminar "Unveiling the mitochondrial architecture of Gastrotricha: insights into evolution and lineage-specific adaptations" - University of Bologna; 06/03/2025
Invited speaker: Anush Kosakyan (University of Modena and Reggio Emilia).

Thesis Co-supervisor

1. (MSc) Giuseppe Nicoletta - Caratterizzazione di sncRNA mitocondriali in *Ruditapes philippinarum* e analisi comparativa tra metazoi (graduated 17/07/2025)
2. (MSc) Alice Cascella - Spiralia Phylogenomics
3. (BSc) Giorgia Galletti - Coevolution of Mitochondrial and Nuclear Proteins in Snakes Revealed by Network-Based ERC Analyses (graduated 28/10/2025)
4. (BSc) Nicolò Gabriele - Analisi della Discordanza Filogenetica Nucleo-Mitocondrio nell'ordine Squamata: Evidenze da Test di Topologia (graduated 28/10/2025)